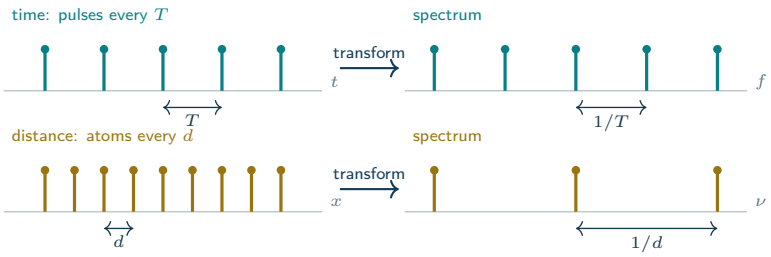
FOURIER ANALYSIS ON ONE PAGE

From t to $1/t$ and from d to $1/d$

The transform pairs a period with a frequency, and a spacing with a spatial frequency.

1 / THE RULE AND THE PICTURE

Pulses spaced T apart and atoms spaced d apart have spikes at multiples of $1/T$ and $1/d$



halving the spacing doubles the spacing of the spikes

The Fourier transform writes a signal as a sum of waves, $F(\omega) = \int f(t) e^{-i\omega t} dt$, and a wave that repeats every T sits at the frequency $f = 1/T$. The same transform applied to a pattern in space, $\tilde{\rho}(k) = \int \rho(x) e^{-ikx} dx$, puts a pattern that repeats every d at the spatial frequency $\nu = 1/d$:

$$T \leftrightarrow \tfrac{1}{T} \qquad d \leftrightarrow \tfrac{1}{d}$$

2 / WHY THE RELATION IS RECIPROCAL

Shrinking a pattern by a factor a stretches its spectrum by the same factor

The scaling rule says that replacing $f(t)$ by $f(at)$ replaces the transform $F(\nu)$ by $\frac{1}{|a|} F(\nu/a)$:

$$f(at) \;\longleftrightarrow\; \tfrac{1}{|a|}\, F\!\left(\tfrac{\nu}{a}\right)$$

Playing a recording twice as fast is $f(2t)$, every frequency doubles, and 440 Hz becomes 880 Hz, an octave higher. In space a lattice with half the spacing has a reciprocal lattice with twice the spacing. For the bump $e^{-\pi(t/\sigma)^2}$ of width σ the transform is $\sigma\, e^{-\pi\sigma^2 f^2}$, of width $\frac{1}{\sigma}$, so the product of the two widths stays fixed, which is the uncertainty bound in the table below.

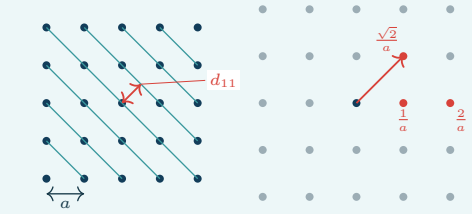
3 / THE TWO TRANSFORMS SIDE BY SIDE

The same mathematics with different units and a few differences

	time	space
variable	time t, in seconds	distance x, in metres
period or spacing	period T	spacing d (or wavelength λ)
frequency	$f = 1/T$ in hertz, with $\omega = 2\pi f$	$\nu = 1/d$ in m^{-1}, with $k = 2\pi\nu$
basic wave	$\cos(2\pi f t)$	$\cos(2\pi\nu x)$
transform	$\int f(t)\, e^{-2\pi i f t}\, dt$	$\int \rho(x)\, e^{-2\pi i \nu x}\, dx$
narrow gives broad	$\Delta t\, \Delta f \geq \frac{1}{4\pi}$	$\Delta x\, \Delta\nu \geq \frac{1}{4\pi}$, that is $\Delta x\, \Delta p \geq \frac{\hbar}{2}$ for $p = h\nu$
sampling step	step T, limit $\frac{1}{2T}$: 44.1 kHz audio reaches 22.05 kHz	step d, limit $\frac{1}{2d}$: $5\,\mu$m pixels reach 100 cycles/mm
real signals	$\lvert F(-f)\rvert = \lvert F(f)\rvert$	$\lvert\tilde{\rho}(-\nu)\rvert = \lvert\tilde{\rho}(\nu)\rvert$ (Friedel's law)
dimensions	one variable	three variables, u a vector, spikes on a reciprocal lattice
what is measured	intensity $\lvert F\rvert^2$, phase lost	intensity $\lvert\tilde{\rho}\rvert^2$, the crystallographic phase problem

4 / FROM d TO $1/d$ IN A CRYSTAL

Bragg's law is the transform of a lattice read at the spatial frequency $2\sin\theta/\lambda$



real lattice and the (11) planes reciprocal lattice, spacing $\frac{1}{a}$

In a crystal the diffraction pattern is the transform of the electron density, so each family of planes with spacing d gives a spot at spatial frequency $1/d$. For a square lattice of side a the planes (h,k) are $d = \frac{a}{\sqrt{h^2+k^2}}$ apart, so the spot lies at $\frac{\sqrt{h^2+k^2}}{a}$ and finer planes sit farther out. Light of wavelength λ scattered through 2θ probes the spatial frequency $\frac{2\sin\theta}{\lambda}$, and requiring this to equal $\frac{1}{d}$ is Bragg's law $\lambda = 2d\sin\theta$. For silicon (111), $a = 5.4309$ Å and copper $K\alpha$ light with $\lambda = 1.5406$ Å, the planes are $d = \frac{a}{\sqrt{3}} = 3.1355$ Å apart, $\frac{1}{d} = 0.3189$ Å^{-1}, $\sin\theta = 0.2457$ and $2\theta = 28.44°$.

5 / TIME AND SPACE TOGETHER

A travelling wave has both frequencies, tied by its speed

A wave $\cos\big(2\pi(\nu x - f t)\big)$ with period T and wavelength λ has the speed $v = \frac{\lambda}{T} = \frac{f}{\nu}$, so in the plane of spatial frequency and frequency it lies on the line $f = v\nu$. The note A_4 at 440 Hz has $T = 2.27$ ms, and in air at 343 m/s the wavelength is 0.780 m, a spatial frequency of 1.283 m^{-1}. Green light of 500 nm has $\nu = 2 \times 10^6$ m^{-1} and $f = 6.0 \times 10^{14}$ Hz. Spectroscopists quote $\frac{1}{\lambda}$ directly as the wavenumber in cm^{-1}, which is 20000 cm^{-1} for this light.

ASIDE / TRY THESE

Silicon (220) has $d = \frac{a}{\sqrt{8}} = 1.9201$ Å, so $\frac{1}{d} = 0.5208$ Å^{-1} and, with the same light, $2\theta = 47.30°$. A Gaussian is its own transform and meets $\Delta x\, \Delta\nu = \frac{1}{4\pi}$ exactly.

LINKS TO OTHER SHEETS

The Euler formula sheet explains the kernel $e^{i\theta}$, and the power series sheet treats the exponential and the logarithm.

Convention. f frequency, ν spatial frequency, $\omega = 2\pi f$, $k = 2\pi\nu$; Δ the standard deviation of the intensity; all numbers were checked by computation.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames